VARJA KRISHNA KOUSHIK

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	5.59
NARAYANA JUNIOR COLLEGE	Class XII (AP STATE BOARD)	2023	79.60%
N. St. Mathew's Public School	Class X (CBSE)	2021	78.40%

EXPERIENCE

Diagonal Matrix Ltd -- Intern | AI Platforms & Data Engineering |

- **Architected a YAML-driven Data Model Accelerator** that translated business requirements into data models, executable SQL DDL schemas, synthetic datasets, and analytics-ready outputs with automated **PostgreSQL ingestion**.
- **Engineered AI-assisted schema generation and validation workflows** covering DDL creation, PK/FK constraint validation, synthetic data generation, and PostgreSQL integration.
- **Built and validated end-to-end SQL/API workflows** across JAI/Langflow, Docker, PostgreSQL, and REST APIs, using **Postman** for API and end-to-end integration testing.

PROJECTS

FlowLens -- Process Intelligence & Counterfactual Transformation Platform | GitHub

- Built a **process-intelligence platform** that reconstructs business workflows from event logs, diagnoses bottlenecks and SLA risk, and evaluates transformation options through **paired counterfactual simulation**.
- Processed **103,853 events** with **4.06 s import, 1.09 s graph analysis, and 7.14 s paired Monte Carlo**, while predictive models achieved **0.9425 ROC-AUC and 0.8219 F1**.
- Unified **process analytics, predictive ML, counterfactual simulation, and business-impact modeling**, validated through **44/44 regression tests and 12/12 acceptance groups**.

CloudMorph -- Enterprise Modernization Graph & Migration Wave Optimizer | GitHub

- Transformed legacy application portfolios into **dependency-aware modernization roadmaps** using **6R strategy assessment, risk/debt/cost analysis, and constraint-based migration planning**.
- Engineered **NetworkX dependency analysis and OR-Tools CP-SAT optimization** under budget, capacity, risk, deadline, residency, compliance, and maintenance constraints, generating **Pareto-efficient migration alternatives**.
- Benchmarked planning workflows across **100-, 500-, and 1,000-application estates**, with scenario analysis and **provider-neutral target architecture** generation.

PlantShift -- Brownfield Smart-Factory Transformation Decision Platform | GitHub

- Built an **Industry 4.0 transformation platform** combining discrete-event simulation, IoT/MQTT telemetry, digital twins, runtime ML, and paired Monte Carlo to prioritize modernization under **budget, effort, risk, and prerequisite constraints**.
- Measured **1,133.92 messages/s in-process telemetry throughput at 1.12 ms p95**, with scenario execution at **8.71 ms p95** and **36 feasible portfolios evaluated in 305.8 ms**.
- Integrated **edge normalization, MQTT, PostgreSQL/TimescaleDB, runtime ML/digital twins, and portfolio optimization** into business-value and transformation-roadmap analysis, validated through **67 backend tests**.

SKILLS

Programming Languages: Python, C++, SQL

AI, Data & Decision Engineering: Machine Learning, Predictive Analytics, Process Mining, Counterfactual Simulation, Constraint Optimization, Scenario Analysis, Sensitivity Analysis, Business Impact Modeling

Tools & Technologies: Microsoft Excel, Power BI, PowerPoint, Git, GitHub, Postman, pytest

Frameworks & Libraries: Scikit-learn, NetworkX, OR-Tools CP-SAT, FastAPI, Pydantic, Pandas, NumPy

Digital Transformation & Solution Design: Application Modernization (6R), Solution Design, Process Intelligence, Industry 4.0, IoT, Digital Twins, Technology Roadmapping

RELEVANT COURSES

Computer Science: Data Structures & Algorithms, Operating Systems, Database Management Systems

Artificial Intelligence & Machine Learning: Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, Computer Vision

Mathematics for AI/ML: Probability & Statistics, Linear Algebra, Discrete Mathematics, Calculus

POSITIONS OF RESPONSIBILITY

Events & Hospitality Coordinator -- Diesta:

Coordinated **event logistics, guest hospitality, and on-ground execution** for institute-level events.

Multimedia Coordinator -- Milan:

Produced and coordinated **digital and multimedia content** for event promotion and participant outreach.

Editorial Coordinator -- Media Council:

Managed **content planning, editorial coordination, and publication workflows** for institute media initiatives.

Souvenir Shop Coordinator -- Alumni Cell:

Coordinated **souvenir product design, inventory, merchandising, and on-site sales** for Alumni Day.

EXTRACURRICULAR

- Represented **Indian Institute of Technology Hyderabad** at the **Inter IIT Sports Meet** as a **Volleyball Player**.
- Competed in multiple **inter-college volleyball tournaments**, representing IIT Hyderabad.